

GRIDBENCH · DESIGN PRECIS

GridBench design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gridbench		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Design made constructable (GBN-DDR-003); mass, stability and cost updated; cost reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices adopted as recommended (GBN-DDR-001); numbers from GBN-CAL-001; deeper clamp, diamond pin, split fence, ballast slabs; priced BOM
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, safety, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

GridBench design precis

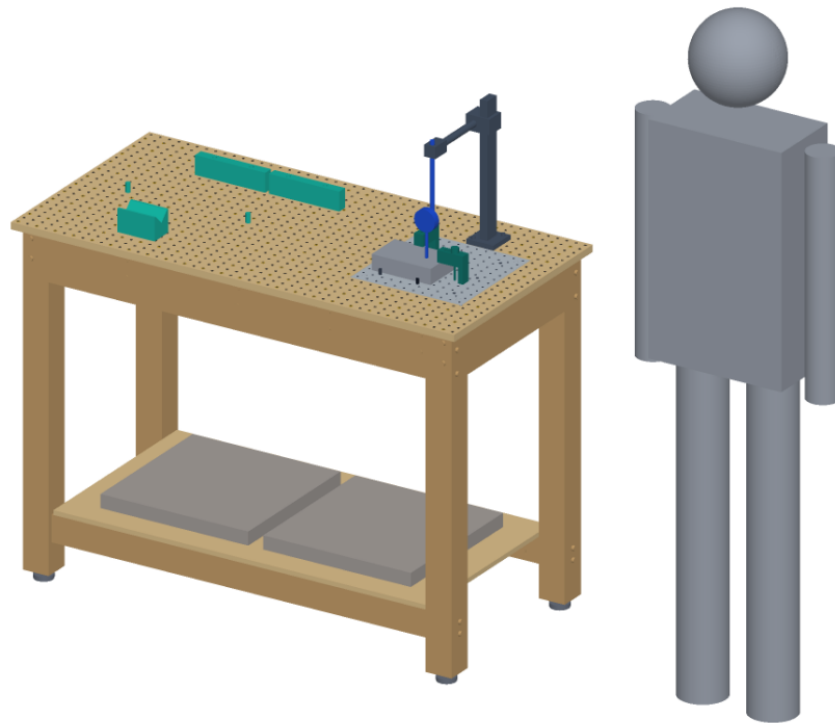
■ Summary

GridBench is a 1,200 x 600 mm workbench whose top is a 25 mm hole grid with M6 threads, the same pattern as metric optical breadboards. A cheap plywood field covers most of the top for general holding, and a 300 x 300 mm aluminum precision tile, set flush on the same grid, carries work that must repeat to a few hundredths of a millimeter. Printed clamps, stops, fences and V-blocks bolt to either zone, and a post with a user-supplied dial indicator checks parts in place. Two concrete slabs on the shelf keep the bench from tipping, and rubber-padded feet keep it from sliding. Value-engineering target: USD 250. Estimated cost of the constructable design: USD 268.40 for the core parts (USD 18.40 over the target). The design was made buildable on 2026-10-01 (GBN-DDR-003): the frame members are bolted to the legs with cross dowels, the worktop is held by brackets and the tile is held by four screws, and every fixture sits on threaded grid positions; the prototype build plan is GBN-BLD-001. The parametric model is `cad/src/model.py`, the general arrangement is drawing GBN-DWG-001 and the calculations are GBN-CAL-001.

GridBench

github.com/BoujeeEnjinia1701/gridbench

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey figure: 1.75 m person for scale

Figure 1. GridBench with a 1.75 m person for scale. Concept, not for fabrication.

■ How it works

1. **One grid everywhere.** Hole centers sit at $12.5 + 25k$ mm from the front-left corner of the worktop, in both directions. The tile's edges fall midway between grid lines, so its holes continue the plywood grid, a fixture can straddle both zones and any fixture file places itself by grid coordinates (column, row).
2. **Locate on dowels, clamp on threads.** As on machinist fixture plates, location and clamping are separate. On the tile, nine reamed 8 mm H7 bores at 100 mm pitch sit in the centers of grid squares, between the M6 holes. A precision fixture or part nest carries one round and one diamond (relieved) 8 mm h6 pin that drop into two bores, then M6 screws clamp it down. Two round pins would bind on any pitch error (GBN-CAL-001, section F). On the plywood field, 6.6 mm plain holes take stop pins, and M6 threads on a 50 mm sub-grid take clamps: flanged tee nuts pressed in from the underside wherever the underside is clear, and screw-in inserts where an apron, rail or leg lies below.
3. **Work and check in the same setup.** The instrument post bolts to any grid hole and carries a dial indicator on an arm, so a part can be checked before it is unclamped (Figure 3).
4. **Fixtures travel as files.** Each fixture is a parametric build123d part with a short metadata block that states its grid footprint and datum. A workshop that builds a GridBench can print or machine a published jig and expect it to fit. Jigs documented with the lab's ReadyKit toolchain can carry their own drawings and revision history.

GridBench: material flow for one part at a fixture (times are estimates)

github.com/BoujeeEnjinia1701/gridbench

CONCEPT, NOT FOR FABRICATION

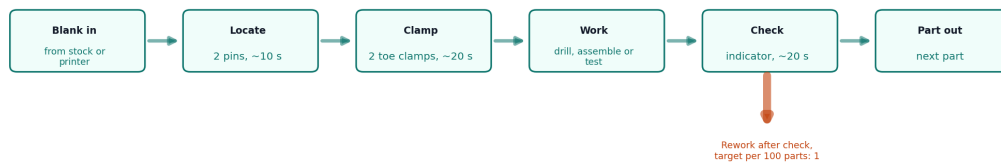


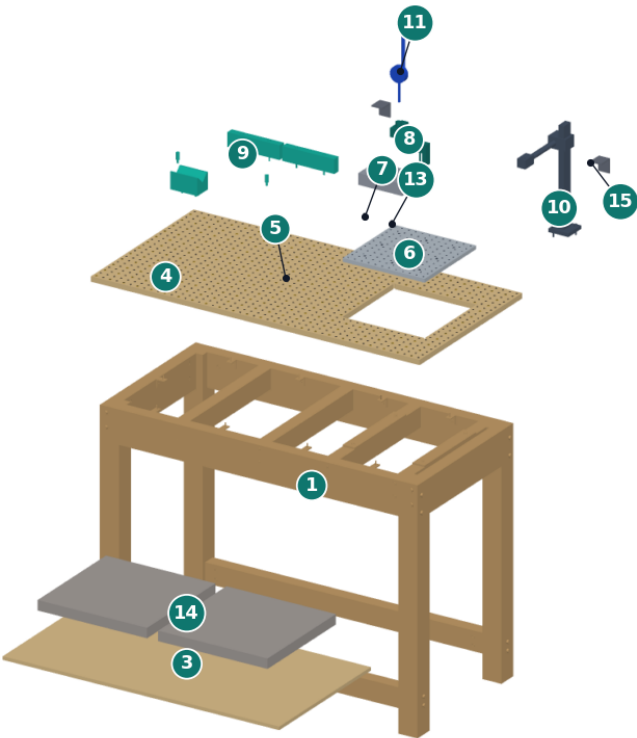
Figure 3. Material flow for one part at a fixture. Times and the rework rate are estimates or targets; the fixture change time is from GBN-CAL-001.

GridBench: exploded view

github.com/BoujeeEnjinia1701/gridbench

CONCEPT, NOT FOR FABRICATION

- 1 Bench frame, bolted timber
- 2 Levelling feet, M10, rubber pads (4) (hidden in this view)
- 3 Lower shelf
- 4 Plywood grid worktop (coarse field)
- 5 M6 tee nuts and inserts, 50 mm sub-grid
- 6 Precision grid tile, aluminum
- 7 Round dowel pin, 8 mm h6
- 8 Printed toe clamps
- 9 Printed fence, stops and V-block
- 10 Instrument post and arm
- 11 Dial indicator, 0.01 mm (user-supplied)
- 13 Diamond locating pin, 8 mm h6
- 14 Ballast slabs, concrete (2)
- 15 Wall anchor brackets (optional)



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 2. Exploded view. Callout numbers match the lines in `bom/bom.csv` .

■ Main components

Table 1. Main components (numbers match the BOM and Figure 2).

No.	Component	Design
1	Bench frame	Bolted softwood: four 70 x 70 mm legs at 1,090 x 490 mm centers, 45 x 120 mm aprons and two low rails butted between the legs on M8 bolts and cross dowels, four 45 x 70 mm cross rails (clear bays 255 mm or less), two 5.3 mm hardwood packers under the tile
2	Levelling feet	Four M10 feet in T-nuts with 3 mm rubber pads, ± 15 mm adjustment
3	Lower shelf	12 mm plywood, 1,010 x 535 mm, resting on the low rails; carries the ballast
4	Plywood grid worktop	18 mm birch plywood, 1,200 x 600 mm, 1,008 holes on the 25 mm grid, 300 x 300 mm through pocket for the tile
5	M6 threaded inserts	252 on the 50 mm sub-grid: 130 flanged tee nuts (19 mm flange, 9.5 mm barrel) pressed in from the underside, and 122 screw-in inserts (10 mm OD by 13 mm) where a frame member lies below
6	Precision grid tile	Cast aluminum tooling plate (offcut), 300 x 300 x 12.7 mm, 144 M6 tapped holes, 9 reamed 8 mm H7 bores, 4 counterbored fixing holes to the two cross rails
7	Round dowel pins	8 mm h6 hardened pins, 20 mm long; one per precision fixture
8	Printed toe clamps	PETG, 70 x 24 x 30 mm body with a 6.6 mm slot and a heel, printed on its side; rated screw torque 1.6 N·m
9	Printed fence, stops and V-block	Fence in two 190 mm segments, each on two grid holes; stop pins for 6.6 mm holes; 90° V-block
10	Instrument post and arm	20 x 40 mm aluminum extrusion on a printed grid foot, 18 mm arm
11	Dial indicator	0.01 mm resolution, 10 mm travel, 8 mm stem; user-supplied
12	M6 fastener kit	Cap screws, washers and printed knobs (no callout)
13	Diamond locating pins	8 mm h6 relieved pins; one per precision fixture, paired with a round pin
14	Ballast slabs	Two 400 x 400 x 35 mm concrete paving slabs, 12.9 kg each, on the shelf
15	Wall anchor brackets	Optional pair of steel angles from the rear apron to a wall
16	Tile machining tools	M6 taps, drills, 8 mm H7 reamer and a tapping guide (no callout)

■ Key numbers

All values are from GBN-CAL-001, where every assumption is stated. They are paper estimates, not test results.

Table 2. Key numbers.

Quantity	Value	Note
Grid positions	1,152 (48 x 24): 144 on the tile, 252 inserts and 756 plain holes	25 mm pitch, 12.5 mm edge offset
Bench mass	43.8 kg empty; 69.5 kg with two ballast slabs	Densities in GBN-CAL-001, Table 1
Tipping force at the top edge	117 N empty; 186 N with the ballast (target 150 N)	Feet 245 mm from the center line, push at 900 mm
Deflection, 500 N at bay center	0.19 mm in the plywood; 0.37 mm including rails and 45 x 120 mm aprons (target 0.5 mm)	Plate theory, E about 7 GPa, pinned joints
Plywood field moisture movement	±0.115 mm over 1,000 mm at 40 to 60 % RH; ±0.40 mm over 20 to 80 % RH	Sourced moisture contents; plywood coefficient 0.007 % per % still unsourced
Tile thermal growth	0.069 mm over 300 mm per 10 K	Aluminum, 23×10^{-6} /K
Dowel relocation	0.036 mm worst case, 0.015 mm statistical (target 0.03 mm)	Round plus diamond pin, 200 mm apart
Clamp force at the part	515 to 858 N at 1.6 N·m; body stress 9.9 MPa or less	M6, nut factor 0.15 to 0.25
Tee nut pull-through	2.69 to 4.30 kN against up to 1.78 kN of screw tension (130 positions)	Assumed shear strength; R7 met on paper
Screw-in insert pull-out	1.02 to 1.63 kN against up to 1.78 kN (122 positions over the frame)	R7 not met there; options await Amish
Sliding force, rubber pads	About 215 N empty, 341 N ballasted	$\mu = 0.5$ assumed
Fixture change	56 s (target 60 s)	Two screws out, two in
Tile machining	About 4.2 h by drill press with a tapping guide	144 holes, 9 reamed bores, 4 counterbores

■ Key design choices

Each choice below is decided by Amish, 2026-09-25: go with recommendation (GBN-DDR-001, GBN-DDR-002).

1. **Grid: 25 mm, M6.** Matches metric optical breadboards, so commercial posts, clamps and bases fit (RP Photonics, "Optical breadboards"). A heavy 50 mm variant is left for later.
2. **Two zones instead of one precise top.** A full aluminum top of 1,200 x 600 mm would cost several times the budget. Plywood gives area; the tile gives precision where it is needed.
3. **Inserts on a 50 mm sub-grid,** with the 25 mm holes kept for pins. Inserts at every plywood hole would add about \$75 and a day of fitting.
4. **Bolted timber frame** for the prototype, with welded steel as a documented variant.
5. **PETG printed fixtures with steel where it matters.** Dowels, screws and the tile are metal; bodies are printed, with a machined option for heavily used jigs.
6. **Tile at the right of the worktop,** leaving the left 800 mm free for large parts.
7. **Ballast on the shelf plus an optional wall anchor,** for stability, and rubber pads on the feet against sliding.

8. **Fixture files carry a metadata block** stating grid footprint and datum; the fixture library stays under CERN-OHL-S-2.0.
9. **Tee nuts from below where the underside is clear**, so clamp load bears on the plywood instead of shearing it; screw-in inserts only over frame members.
10. **45 x 120 mm aprons**, giving the worktop a margin on R6 (0.37 mm against 0.5 mm).

Four details changed at TRL 3 because of GBN-CAL-001 v0.1, within these choices: the toe clamp body is 30 mm deep instead of 16 mm and its rated torque is 1.6 N·m instead of 2 N·m (the TRL 2 clamp would have carried about 33 MPa and crept); each precision fixture uses one round and one diamond pin; the fence is two 190 mm segments so that it fits a 200 mm print bed; and the ballast is two concrete slabs, 25.8 kg in all.

■ Cost

Value-engineering target: USD 250 (`budget_usd` , a hypothetical control target, not a limit; Amish, 2026-10-01). Estimated cost of the constructable design: USD 268.40 for the core parts (USD 18.40 over the target), from the priced BOM (`bom/bom.csv` , 17 lines), with the dial indicator (\$15.00) user-supplied and the wall anchor brackets (\$5.00) optional; everything together is \$288.40. The largest items are the frame (\$57.00, of which \$23.40 is bolts, cross dowels and screws), the plywood worktop (\$45.00), the threaded inserts (\$28.20) and the tile offcut (\$28.00, unverified). The joints, fixings and clamps that make the design buildable (GBN-DDR-003) added \$22.50 to the \$245.90 of GBN-CAL-001 v0.2. The main cost drivers and the savings worth trying are in the design decisions register (GBN-DEC-001).

■ Safety

Safety: Drilling, tapping and routing the worktop and tile create chips and projectiles: wear eye protection and clamp work before cutting. Clamps and toe clamps create pinch points; keep fingers clear when tightening. The empty bench tips at about 117 N at the top edge (GBN-CAL-001), below the 150 N design push, so place both ballast slabs on the shelf or anchor the bench to a wall before use, and lift each 13 kg slab with a straight back. Chamfer or round all plywood, tile and printed edges. Printed clamps can crack and release a part under load; do not exceed the rated screw torque of 1.6 N·m, and do not use printed clamps for machining with a milling machine. Screw-in inserts over the frame members may pull out before a clamp reaches its rated force; until the light-duty rating is decided, use the tee-nut positions or the tile for full clamping force. The bench is not for welding, grinding sparks or hot work: plywood and PETG are combustible.

■ Open questions

- [] R7 at the 122 screw-in positions over the frame: light-duty rating at 0.92 N·m (recommended), moving the sub-grid, or a relaxed target (GBN-DDR-002 O4). Proposed, awaiting Amish.
- [] Racking stiffness: end low rails or a screwed-down shelf (GBN-DDR-002 O3). Proposed, awaiting Amish.
- [] First workshops and regions for co-design (GBN-DDR-001 O1). Proposed, awaiting Amish.
- [] Can the tile be drilled to ± 0.05 mm without CNC, using a printed or steel drill jig on a drill press? Not verifiable on paper.
- [] A sourced figure for plywood in-plane moisture movement; the moisture contents are now sourced, the coefficient is not.
- [] Flatness of cast tooling plate bought as offcut, and of the plywood sheet once screwed down (R5).